

Imbalanced Vehicle Classification

Phase 1 — Preliminary Report

Introduction to Artificial Intelligence — Summer 2026

May 2026

1. Dataset Description

1.1 Overview

The dataset used in this project originates from the Multi-Modal Aerial View Object Classification Challenge (PBVS 2022). It contains aerial images of vehicles categorized into 10 classes. The training set consists of 285,772 images in total, while the validation set contains 770 images (77 per class, perfectly balanced).

The dataset presents a significant class imbalance challenge in the training split. Sedans alone account for approximately 82% of all training samples, while the rarest classes (Bus, Flatbed truck w/ trailer) represent less than 0.25% each.

1.2 Class Distribution

Class	Vehicle Type	Train Samples	Validation	Train %
0	Sedan	234,209	77	82.0%
1	SUV	20,089	77	7.0%
2	Pickup truck	15,301	77	5.4%
3	Van	10,655	77	3.7%
4	Box truck	1,741	77	0.6%
5	Motorcycle	852	77	0.3%
6	Flatbed truck	828	77	0.29%
7	Bus	624	77	0.22%
8	Pickup truck w/ trailer	840	77	0.29%
9	Flatbed truck w/ trailer	633	77	0.22%

1.3 Key Observations & Findings

- Extreme imbalance: the Sedan class (234,209 samples) has ~374x more samples than the Bus class (624 samples). This is a severe long-tail distribution.
- The 8 minority classes (classes 2-9) together represent only ~10.8% of the training data.

- The validation set is perfectly balanced (77 samples per class), meaning a model that overfits to the majority class would show poor validation performance.
- Visual inspection of aerial images shows that some classes are visually very similar (e.g. Pickup truck vs Pickup truck w/ trailer), which is an additional difficulty beyond the imbalance.
- Image resolution and viewpoint vary across samples since images are captured from aerial/drone perspectives, which may affect feature extraction.

2. Proposed Solution Plan

2.1 Dataset Splitting Strategy

Since the validation set has no labels, a portion of the training data will be reserved as an internal validation/evaluation split. The strategy is as follows:

- Apply a stratified split to preserve class proportions: 80% of training data for model training, 20% for internal validation.
- Stratification ensures that even minority classes are represented in both splits.
- The official unlabeled validation set (770 images) will be used for final prediction submission.

2.2 Algorithms to be Used

We plan to use a Convolutional Neural Network (CNN)-based approach, leveraging transfer learning from a pretrained backbone:

- Primary model: ResNet-50 pretrained on ImageNet, fine-tuned on the vehicle dataset. ResNet-50 provides a good balance between accuracy and computational efficiency.
- Alternative/comparison model: EfficientNet-B0, known for strong performance on image classification with fewer parameters.
- Both models will be fine-tuned end-to-end with a custom classification head (10 output classes).

2.3 Class Imbalance Strategy

We will experiment with and compare the following strategies to address class imbalance:

- Weighted loss function: assign higher loss weights to minority classes (inversely proportional to class frequency). This is the primary approach.
- Oversampling with augmentation: use RandomOverSampler or custom weighted sampling (WeightedRandomSampler in PyTorch) to balance the batch composition during training.
- Data augmentation: apply aggressive augmentation (random flips, rotations, color jitter, random crops) especially on minority classes to increase their effective diversity.
- Class-balanced batching: ensure each training batch contains at least a minimum number of samples from each class.

2.4 Main Tools, Frameworks and Libraries

- Deep Learning: PyTorch (model training, dataloaders, loss functions)
- Pretrained Models: torchvision (ResNet-50, EfficientNet-B0)
- Data Processing: NumPy, Pandas, Pillow (PIL)
- Visualization: Matplotlib, Seaborn (training curves, confusion matrix, sample plots)
- Imbalanced Learning: PyTorch WeightedRandomSampler, custom loss weighting
- Evaluation: scikit-learn (classification report, confusion matrix, F1-score)
- Version Control: GitHub repository for code and report

2.5 Evaluation Methods

Given the class imbalance in training and the balanced validation set, standard accuracy alone would be misleading. The following evaluation metrics will be used:

- Per-class accuracy and macro-averaged accuracy: averages accuracy equally across all 10 classes.
- Macro F1-score: harmonic mean of precision and recall averaged over all classes — the primary metric given the imbalance context.
- Confusion matrix: visualized as a heatmap, to identify which classes are most confused with each other and detect systematic prediction errors.
- Per-class Precision, Recall and F1-score: detailed classification report to understand performance on minority classes specifically.
- Training/validation loss and accuracy curves: to monitor overfitting and convergence behavior.

The confusion matrix is particularly important in this project because the validation set is balanced — it will clearly reveal if the model is biased toward predicting Sedan (the majority class) regardless of the actual input.

3. References

[1] S. Low et al., "Multi-modal Aerial View Object Classification Challenge Results - PBVS 2022," CVPRW 2022.

[2] <https://codalab.lisn.upsaclay.fr/competitions/1392>

[3] J. M. Johnson and T. M. Khoshgoftaar, "Survey on deep learning with class imbalance," Journal of Big Data, vol. 6, no. 1, p. 27, 2019.

[4] <https://towardsai.net/p//multi-class-model-evaluation-with-confusion-matrix-and-classification-report>